D72 — Interface Asymmetry of the Proton (Version 2)

The $(24, 24, 28)/76$ core–sea interface, the spin correction, and the parity-defect reading of the valence spin budget

PDL Programme — laubscher-lab
`https://github.com/laubscher-lab/PDL-framework`

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Version note (v2). This version supersedes D72 v1 (Zenodo DOI 10.5281/zenodo.23783579, 29 October 2026) [4]. Version 1 modelled the interface pattern $(28, 28, 24)$ over $I_{\text{tot}} = 80$ under the proton label. As recorded in the programme context and DOI registry [1], that pattern is the *neutron* interface; the proton interface is $(24, 24, 28)$ over $I_{\text{tot}} = 76$, with the distinguished d core the *long* core. Every consequence is propagated below (campaigns v19–v20). The exact result $\Delta\Sigma = 24/80 = 3/10$ of v1 is not lost: it is re-attributed to the neutron motif, where it is arithmetically admissible (V19-H). What changes for the proton is the spin budget and the interface-forcing argument; the leakage derivation itself is untouched.

Epistemic labels. (T) = theorem forced inside the model; (D) = derived within an a priori declared family (forms, denominators and integer shifts declared before comparison); (H) = heuristic/compatibility statement. Nothing in this note claims a derivation of measured quantities from first principles; see Section 6.

1 Architectural base values (unchanged)

The architectural constants of the corpus are unchanged by the v2 correction [3, 6]. The parity of the sea, $11017 = 23 \times 479$ (odd), forces a conserved mobile monomer defect in any relational closure of the sea (T); under the surface ontology of D71 (no core interior, K_4 bows, the sea as the unique odd graph), the wake census is a boundary count.

Quantity	Value	Status
Coherence units	$n_u = 930 = 918 + 12$	corpus
Sea	918 sites, $11017 = 23 \times 479$ relations (odd)	corpus
Parity monomer	unique conserved mobile defect	(T)
Valence cores	$3 \times K_4$: 12 core sites, 18 internal relations	corpus (RN1: $R_e = 6 \neq 24, 28$)
Bow degree	$d_{\text{bow}} = 8 = 24/3$	(D)
Wake census	$B = 528 = 24 \times 22$	(D)
Diagonal cut	$\sqrt{2}/8$ per locus; $B = 528 + 1/(2\sqrt{2})$	(D)
Closed fraction	$f = 144/917 = 12^2/(918 - 1) = 0.157126$	(D)

2 The corpus interface counts: proton versus neutron

The corpus assigns interface entity counts per valence core. Proton: $n_K = n_u + n_u + n_d = 24 + 24 + 28 = 76$. Neutron: $28 + 28 + 24 = 80$. Registry line: "76 (proton, $2 \times K_{24} + K_{28}$), 80 (neutron, $K_{24} + 2 \times K_{28}$)" [1]. In the proton the distinguished d core is therefore the *long* core (28 entities). The isospin shift between the two baryons is $80 - 76 = 4$, exactly one core-site unit (sites/core $= 12/3 = 4$).

Version 1 used $I_{\text{tot}} = 80$ with the split $(28, 28, 24)$ — the neutron pattern — under the proton label. On that (wrong-for-proton) total, two criteria (deficit = sites/core = 4; short core divisible by $d_{\text{bow}} = 8$) selected $(28, 28, 24)$ uniquely among the declared 2+1 partitions of 80 (V18-A), with weighted wake shares 184.8/184.8/158.4 summing to 528. On the correct proton total 76 the same criteria do *not* select a unique split: **(V19-A, FAIL)** the interface-forcing argument must be rebuilt for the proton and is registered as open problem OP-D72-1. This failure does not propagate to the counts themselves, which are corpus data, not fitted parameters.

3 What the correction leaves intact (campaign v19)

(V19-B1, PASS) Interface weights $\omega_c = I_c/I_{\text{tot}} = \{6/19, 6/19, 7/19\}$; the weighted wake shares sum exactly to $B = 528$, and the symmetric share $76/3$ is non-integer — an asymmetric split remains required for proportional bookkeeping.

(V19-B2, PASS) On the declared family $x(x-2)$, the unique hit is $24 \times 22 = 528$: the wake census re-reads on a u interface ($I_u = 24$), structurally coherent with $B = n_u(n_u - 2)$ where n_u denotes the u-core interface count 24.

(V19-C, PASS) Predicted locus degrees $\{I_c + 1\} = \{25, 25, 29\}$; the two degree-25 loci located in v13 therefore sit on *a u core*; the spin reference bow is a u bow.

Leakage chain (T1), intact. Parity (T) $\to$ monomer $\to$ wakes $B = 528 \to$ diagonal cut $\sqrt{2}/8$ per locus $\to B = 528 + 1/(2\sqrt{2}) \to$ correction $\delta = 1/\binom{8}{3} = 1/56 \to \Delta r_{\text{pred}} = 0.04795965$ versus $\Delta r_{\text{val}} = 0.04795969 \pm 7 \times 10^{-8}$, residual $4.5 \times 10^{-8} = 0.643$ tol (PASS). None of these steps involves the u/d assignment of the interface.

(V19-E, PASS) The diagonal-cut fraction $1/(2\sqrt{2})$ survives the interface audit: no interface rational falls inside its window — its origin is geometric, not bookkeeping. **(V19-F, PASS)** $f = 144/917$ counts core sites, never interface links: it is immune to the asymmetry.

4 The spin budget: an integrality obstruction and its parity reading (campaign v20)

4.1 The exact v1 value migrates to the neutron

$\Delta\Sigma = I_{\text{dist}}/I_{\text{tot}} = 24/80 = 3/10$, residual 0.000 against the EMC window [8], is verified-as-neutron (V19-H): it is the reading of the neutron motif $(28, 28, 24)/80$ with a distinguished core of interface 24. For the proton the naive readings fail twice: $28/76 \approx 0.368$ (outside the window) and $24/76 \approx 0.316$ (residual $\approx 5\%$).

4.2 Integrality obstruction

The proton cannot carry 3/10: $76 \times 3/10 = 114/5 \notin \mathbb{Z}$. The nearest admissible numerator is 23:

$$\frac{23}{76} = 0.302632, \qquad \left|\frac{23}{76} - \frac{3}{10}\right| = \frac{1}{380} = \frac{1}{5\, I_{\text{tot}}}.$$

The gap is closed-form, not noise.

4.3 The dual identity of the numerator 23

(a) $23 = I_u - 1$: the u interface minus the unique conserved monomer forced by the sea's parity (T). (b) 23 is the prime parity carrier of the sea itself: $11017 = 23 \times 479$. The same integer governs the parity of the sea and the spin budget of the proton.

4.4 Physical reading

$$\Delta\Sigma_{\text{proton}} = \frac{I_u - 1}{I_{\text{tot}}} = \frac{23}{76},$$

i.e. the proton's spin reads the u interface *corrected by the parity defect*. The same monomer that governs the leakage (T1) and the closed fraction (T2) also strikes the spin (T3), shifting $24/76 \to 23/76$ and moving the proton exactly $1/380$ away from the neutron's exact $3/10$. Three targets, one defect. The isospin partner reads are: neutron — exact form $3/10 = 24/80$ on the distinguished core; proton — defect-corrected form $23/76$ on the u interface, consistent with the u-core localisation of the loci (V19-C).

4.5 Status and multiplicity control

The candidate $23/76$ carries label (D) within the a priori declared family "structural integer $\pm k$, $k \leq 2$" applied to the distinguished interface. Honest control: the EMC window is wide — 8 fractions $a/76$ fall inside it, against a mean expectation of ≈ 7.6; window proximity alone is statistically ordinary and proves nothing. The candidate is therefore carried by the *dual structural identity of the numerator* (Section 4.3), not by the window. It improves T3 from $7/24$ (residual 0.167, built on interchangeable cores) to residual 0.0526 — a factor ≈ 3.2 — and $7/24$ remains recorded in the corpus as the superseded symmetric-chain form.

5 Unified audit

Target	Observable	Value	Residual (tol)	Verdict
T1	proton skin Δr	0.04795965 vs $0.04795969 \pm 7 \times 10^{-8}$	0.643	PASS
T2	closed fraction f	$144/917 = 0.157126$	0.102	PASS
T3	valence spin $\Delta\Sigma$	$23/76 = 0.302632$	0.0526	PASS

Triple PASS without any exact target for the proton: exactness (3/10) has migrated to the neutron motif $(28, 28, 24)/80$ (V19-H) — a proton↔neutron symmetry of the mechanism that warrants its own campaign (OP-D72-2, candidate D73).

6 Limitations, falsifiers, open problems

(i) All second-generation claims carry status (D) within a priori declared families — not theorems. (ii) The T1 fractional signature remains at $\sim 1.7\sigma$. (iii) The T2 hits are correlated. (iv) The EMC spin window is wide; the T3 multiplicity control of Section 4.5 is engraved. (v) The circularity of G is NOT lifted: the continuous chain C1–C4 $\to B \to \Delta r_{\text{val}} \to \varepsilon_G \to G_{\text{PDL}}$ remains to be written (OP-D71-1). (vi) OP-D72-1: the interface-forcing argument for the proton total 76 is

open (V19-A). (vii) OP-D72-2: the neutron campaign around the exact 3/10 is open. The pre-registered falsifier stands unchanged: any admissible boundary law must reproduce simultaneously $\Delta r_{\mathrm{val}} = 0.04795969 \pm 7 \times 10^{-8}$ and $f = 0.157126$ from one law.

7 Reproducibility

Scripts (standard library only): v17 — interface hypothesis audit; v18 — analysis on the 80-pattern (as published in v1); v19 — proton correction (nine checks); v20 — spin-gap reading. Scripts and JSON summaries ship with the programme archives [7]; corpus documents and DOIs in [1, 2, 3, 4, 5, 6].

References

[1] PDL Programme, *PDL_context.md — programme context and state* and DOI registry *10.5281zenodo.txt*, laubscher-lab/PDL-framework, GitHub (2026). `https://github.com/laubscher-lab/PDL-framework`

[2] PDL Programme, *D70*, Zenodo (2026). DOI: 10.5281/zenodo.23775498.

[3] PDL Programme, *D71 — Parity-driven proton leakage*, Zenodo (2026). DOI: 10.5281/zenodo.23776625.

[4] PDL Programme, *D72 v1 — Interface asymmetry of the proton*, Zenodo (2026). DOI: 10.5281/zenodo.23783579. Superseded by the present version.

[5] PDL Programme, *PDL Global Mapping, version 34 (DM v34)*, Zenodo. DOI: 10.5281/zenodo.22067627.

[6] PDL Programme, *PDL Global Mapping of Structures, Results, and Open Problems, version 36 (DM36)*, Zenodo (2026). DOI: 10.5281/zenodo.23784141.

[7] PDL Programme, *Proton leakage campaign scripts v6–v20* (standard library only, Colab-ready), reproducibility archive, laubscher-lab/PDL-framework (2026).

[8] J. Ashman *et al.* (European Muon Collaboration), Phys. Lett. B **206** (1988) 364; Nucl. Phys. B **328** (1989) 1.